

Rationalized Finite-Adele Row

Autonomous discussion Pass 97 lifts the compact local-Loebification comparison to the rationalized finite-adele skeleton.

For finite support S , the rationalized map of two-term complexes is

$$[\mathbb{Q} \rightarrow \prod_{p \in S} \mathbb{Q}_p] \rightarrow [\mathbb{Q}^S \rightarrow \prod_{p \in S} \mathbb{Q}_p],$$

diagonal in degree 0 and identity in degree 1. The induced map on H^1 is

$$(\prod_{p \in S} \mathbb{Q}_p) / \Delta \mathbb{Q} \rightarrow \prod_{p \in S} (\mathbb{Q}_p / \mathbb{Q}).$$

Its kernel is

$$K_{\mathbb{Q},S} = \mathbb{Q}^S / \Delta \mathbb{Q}.$$

Thus rationalization does not kill the compact kernel

$$K_{\mathbb{Z},S} = \mathbb{Z}^S / \Delta \mathbb{Z}.$$

It embeds $K_{\mathbb{Z},S}$ into a divisible $\mathbb{Q}$ -vector boundary of dimension $|S| - 1$.

The finite shadow is regraded. Since $K_{\mathbb{Q},S}$ is divisible,

$$K_{\mathbb{Q},S} / N K_{\mathbb{Q},S} = 0.$$

But

$$K_{\mathbb{Q},S} / K_{\mathbb{Z},S} \cong (\mathbb{Q}/\mathbb{Z})^S / \Delta(\mathbb{Q}/\mathbb{Z})$$

has N -torsion of size

$$N^{|S|-1}.$$

So the old finite shadow survives as torsion in the rational/integral boundary quotient, not as a finite quotient of the rational kernel itself.

Support projections remain surjective after rationalization. If $S \subseteq T$, the projection

$$K_{\mathbb{Q},T} \rightarrow K_{\mathbb{Q},S}$$

has kernel of $\mathbb{Q}$ -dimension $|T| - |S|$, so the support direction remains Mittag-Leffler.

The checker `artifacts/reports/pass97-rationalized-finite-adele-row-check.json` verifies the rational kernel dimensions, finite-shadow regrading, exact sequence, and support-projection behavior.